Task 2 – Feature Engineering, Model Optimization & Performance Comparison

AI & Machine Learning Internship

Submitted By

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Task: Task 2

Project Title: House Price Prediction using Multiple Regression Models

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1. Introduction

Machine learning models often require proper data preprocessing and feature engineering to achieve better performance. This project focuses on improving a house price prediction model by applying feature scaling and comparing multiple regression algorithms.

The project uses the California Housing dataset provided by Scikit-learn. Different regression models are trained and evaluated to identify the model that provides the most accurate predictions.

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2. Objective

The main objectives of this project are:

- To understand the California Housing dataset.
- To perform data preprocessing and feature scaling.
- To train multiple regression models.
- To evaluate model performance using different metrics.
- To compare the models and identify the best-performing one.
- To save the trained model for future use.

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3. Tools and Technologies Used

The following tools and libraries were used during the project:

- Python
- Jupyter Notebook
- Visual Studio Code
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib

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4. Dataset Description

The California Housing dataset is available in the Scikit-learn library.

The dataset contains information collected from various housing districts in California. It includes several numerical features such as:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

The target variable is the median house value.

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5. Project Workflow

The project was completed using the following steps:

Step 1

Imported all required Python libraries.

Step 2

Loaded the California Housing dataset into a Pandas DataFrame.

Step 3

Explored the dataset by checking:

- Shape
- Data types
- Statistical summary
- Sample records

Step 4

Checked the dataset for missing values and duplicate records.

Step 5

Visualized the data using:

- Histograms
- Correlation Heatmap
- Scatter Plot
- Box Plot

Step 6

Separated the input features and target variable.

Step 7

Applied StandardScaler to normalize the feature values.

Step 8

Split the dataset into training and testing sets using an 80:20 ratio.

Step 9

Trained the following regression models:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Step 10

Generated predictions for the testing dataset.

Step 11

Evaluated each model using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Step 12

Compared all models and selected the best-performing one.

Step 13

Saved the best model using Joblib.

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6. Models Used

Linear Regression

Linear Regression establishes a linear relationship between the input features and the target variable. It is simple, fast, and widely used for regression problems.

Ridge Regression

Ridge Regression is an extension of Linear Regression that applies regularization to reduce overfitting and improve model stability.

Decision Tree Regressor

Decision Tree Regression predicts continuous values by splitting the dataset into multiple decision nodes. It is capable of capturing more complex relationships between features.

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7. Evaluation Metrics

The performance of each model was evaluated using the following metrics:

Mean Absolute Error (MAE)

Measures the average difference between actual and predicted values.

Root Mean Squared Error (RMSE)

Measures the prediction error while giving higher importance to larger errors.

R^2 Score

Indicates how well the model explains the variance in the target variable.

A higher R^2 Score generally indicates better performance.

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8. Results

All three regression models were trained and evaluated successfully.

The evaluation metrics were compared to determine the most suitable model for house price prediction.

The model with the highest R^2 Score and the lowest RMSE was selected as the final model.

The selected model was saved as:

best_model.pkl

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9. Outcome

This project helped in understanding the importance of feature engineering and model comparison in machine learning.

By applying feature scaling and comparing multiple regression algorithms, it was possible to identify a model that performed better for the given dataset.

The project also provided practical experience in preprocessing, visualization, model evaluation, and saving trained models for future use.

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10. Conclusion

This task demonstrated the complete workflow of a machine learning regression project, starting from data preprocessing to model evaluation and selection.

Feature scaling improved data consistency, while comparing different regression algorithms helped identify the most suitable model for predicting house prices.

Overall, this project strengthened my understanding of machine learning model development, evaluation techniques, and practical implementation using Python and Scikit-learn.
